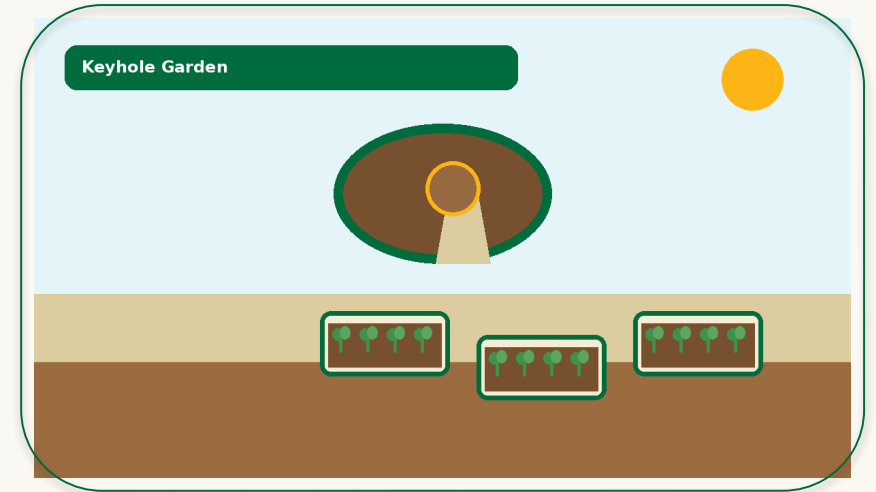


Keyhole Garden

Circular Food Production Systems

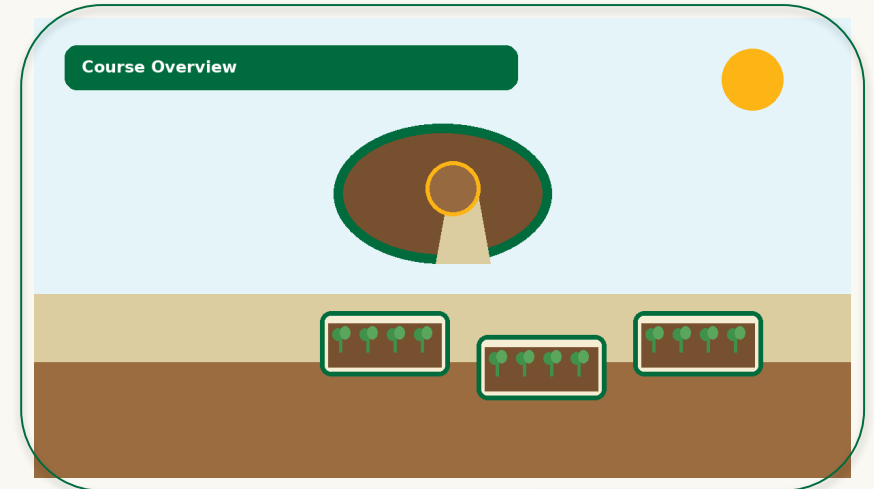
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Course emphasis: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden system and its educational purpose; teach the ten technical topics at Primary 4-6 level; connect mathematics/science/agriculture to practical work; supervise measurements and garden tasks safely; maintain usable records; guide pupils from observation toward evidence-based conclusions.
- Core materials: teacher guide, crop plan, child-safe tools, watering equipment, labels, clipboards, rulers/tapes, scales or thermometers where relevant, observation/data sheets, clean containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson/measurement or investigation plan.



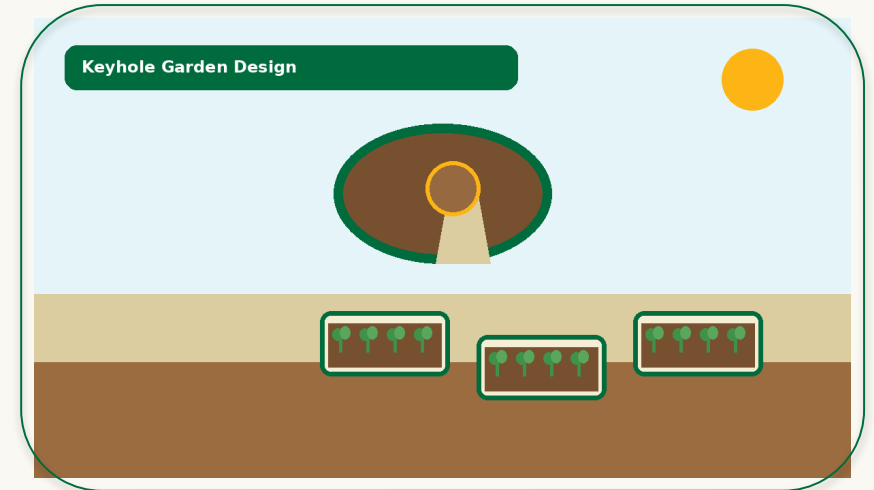
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

Keyhole Garden Design

Teacher knowledge and practical application

- Identify the circular or near-circular growing bed, access notch and central basket typical of a keyhole design.
- Explain how the access notch allows the gardener to reach the center without stepping into the bed.
- Discuss the design as one option for intensive school gardening, not a universal solution.
- Check reach distances and safe edge heights.
- Use a simple plan view to teach system layout.



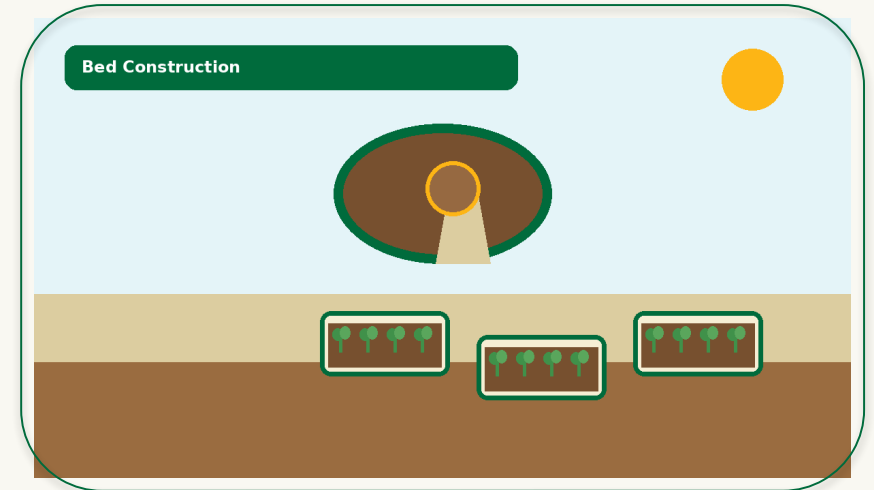
TEACHER PRACTICE

- Prepare a labeled keyhole-garden diagram.

Bed Construction

Teacher knowledge and practical application

- Construction is adult-led where heavy materials, cutting or structural work is involved.
- Use stable, non-hazardous edging materials suitable for the school site.
- Build on a prepared, drained location.
- Ensure the access notch and paths are wide and stable.
- Inspect the completed structure before pupil use.



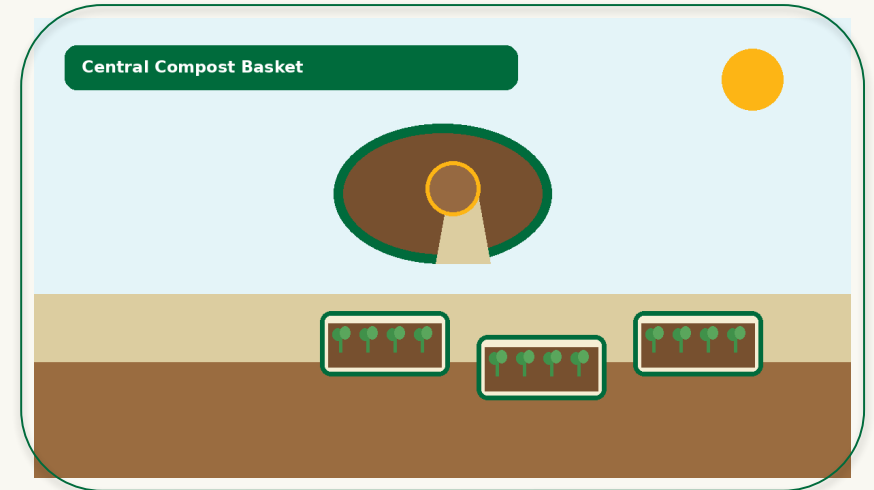
TEACHER PRACTICE

- Teacher check: SITE • STRUCTURE • ACCESS
- DRAINAGE • EDGES.

Central Compost Basket

Teacher knowledge and practical application

- Explain the central basket as a location for approved compostable material in some keyhole designs.
- Use only materials permitted by the school compost protocol.
- Keep fresh manure, meat, hazardous waste and contaminated materials out.
- Ensure the basket is stable and inaccessible to unsafe rummaging.
- Explain that nutrient and water movement from the basket can vary with design and management.



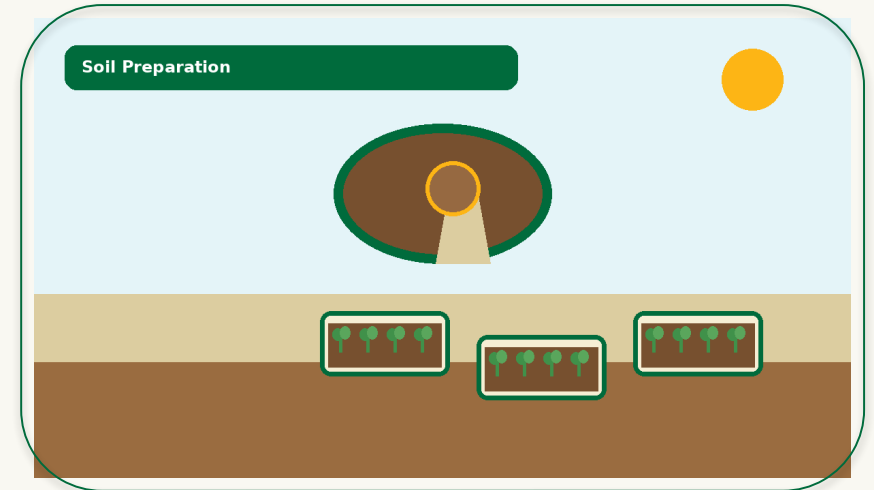
TEACHER PRACTICE

- Prepare a safe-material sorting activity.

Soil Preparation

Teacher knowledge and practical application

- Fill the bed with clean soil/growing media and mature amendments suitable to crop production.
- Avoid assuming all layered construction recipes perform identically in every site.
- Level planting zones while maintaining the intended bed profile.
- Check drainage before planting.
- Keep pupils away from heavy filling operations.



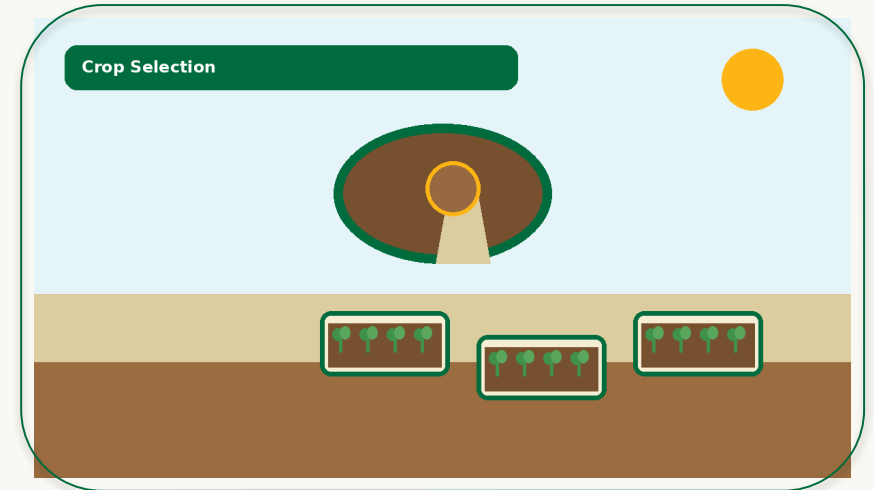
TEACHER PRACTICE

- Prepare the final surface for a supervised planting lesson.

Crop Selection

Teacher knowledge and practical application

- Choose crops suited to bed depth, season and school maintenance capacity.
- Arrange crops so taller plants do not unnecessarily shade smaller ones.
- Keep frequently harvested crops accessible.
- Avoid overcrowding near the central basket or edges.
- Use locally appropriate food crops.



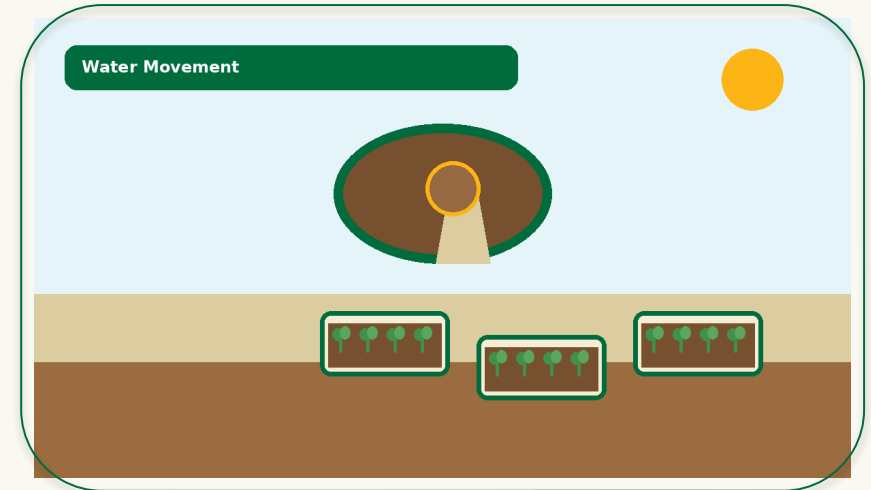
TEACHER PRACTICE

- Create a simple crop-placement map.

Water Movement

Teacher knowledge and practical application

- Observe how water is applied to the bed and central basket if used.
- Explain that water moves through soil under gravity and capillary processes in simplified terms.
- Check different bed positions for moisture rather than assuming uniform distribution.
- Avoid overwatering the center.
- Keep runoff from the access path.



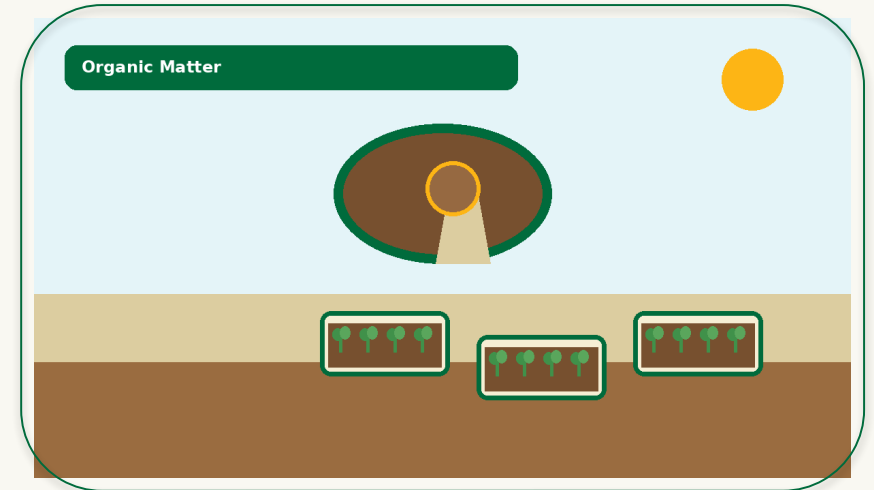
TEACHER PRACTICE

- Set up a center-versus-edge moisture observation.

Organic Matter

Teacher knowledge and practical application

- Explain organic matter as decomposed or decomposing plant/animal-derived material in soil systems.
- Use mature compost as the main pupil-safe example.
- Discuss how organic matter can influence soil structure and water retention.
- Avoid promising exact fertility effects without measurement.
- Apply approved materials at appropriate rates.



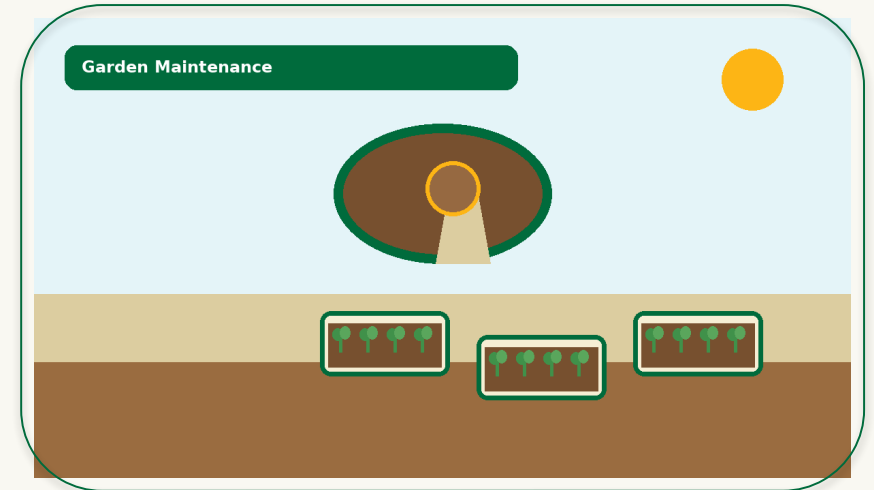
TEACHER PRACTICE

- Compare soil with and without mature compost visually and by feel.

Garden Maintenance

Teacher knowledge and practical application

- Maintain bed edges, access notch, basket, soil surface, irrigation and crops.
- Remove weeds and damaged material using safe procedures.
- Monitor settlement and top up approved media when necessary.
- Keep the central basket managed rather than overloaded.
- Record repairs and recurring moisture problems.



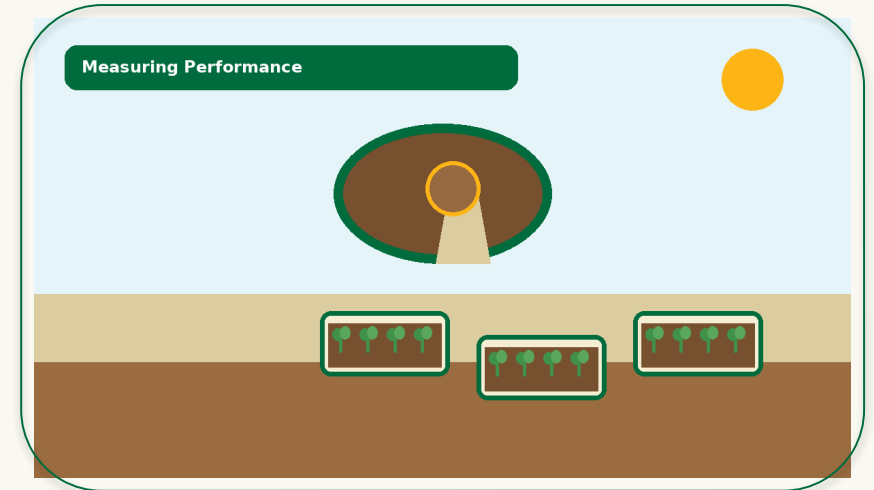
TEACHER PRACTICE

- Create a weekly keyhole maintenance checklist.

Measuring Performance

Teacher knowledge and practical application

- Track crop survival, growth, harvest and selected moisture observations.
- Compare center, middle and outer positions only with consistent methods.
- Record dates and units.
- Use results to identify management questions rather than assume design effects.
- Include maintenance effort as a practical performance consideration.



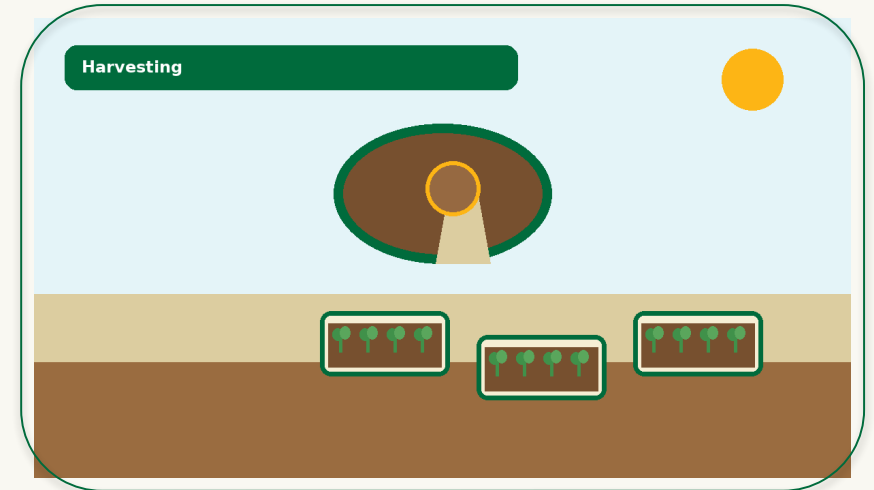
TEACHER PRACTICE

- Prepare a three-zone monitoring sheet.

Harvesting

Teacher knowledge and practical application

- Harvest from the access notch and perimeter without stepping into the bed.
- Use crop-specific readiness indicators.
- Use clean containers and approved tools.
- Record quantity by crop and, if useful, bed position.
- Follow school food-safety procedures.



TEACHER PRACTICE

- Completion task: plan a keyhole-garden crop cycle and monitoring activity.